The Parkinson's Disease Cognitive Composite: Lessons Forgotten from Preclinical Alzheimer's Cognitive Composites

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We were intrigued by Weintraub and colleagues'[1] creation of a Parkinson's disease cognitive composite (PDCC). As the authors emphasize, the goal of composites, exemplified by the Preclinical Alzheimer's Cognitive Composite (PACC), is to create a psychometrically sound index, superior in detecting and tracking disease processes for clinical trials relative to constituent scores. We agree that a composite is needed in Parkinson's disease (PD), but it requires key validation steps before the serious adoption of such applications can be considered: (1) theoretic or empiric justification behind construction; and (2) evaluation of the PDCC's reliability and meaningful clinical validation using biomarker or longitudinal outcomes.

In PD, there is neither consensus around, nor widespread availability of, a biomarker as there is in Alzheimer's (AD) (eg, amyloid positivity). Further, PD has a panoply of outcomes, with cognitive decline being less prominent relative to AD. PACC construction relied on theoretical, data driven, and psychometric considerations.[2,3] For example, PACCs were developed from the cognitive domains/tests known to decline earliest in AD and were determined to have reliability, structural validity, predictive validity, and to be sensitive to change.[2,4,5] In contrast, this PDCC's test selection was not linked to predictive measures of PD cognitive decline.[6] Instead, original Parkinson's Progression Markers Initiative (PPMI) battery tests were included without theoretic justification, and subsequent PPMI additions (eg, letter fluency) that may better track or predict progression were excluded. Further, the authors' largest effect size was in a constituent task (Symbol Digit Modalities) and not the PDCC. As presented in the paper, the PDCC differs between medication naive PD patients and controls, but clinically, the PD group's z-score is within normal limits. More robust validation and clarity on other methodological ambiguities (eg, practical application of robust norms, addressing skewed residuals) are needed to support the PDCC. By not considering the concerns raised above, the utility of the composite is limited and not inherently specific to PD.

To determine whether the proposed PDCC has reliability and structural validity, we reanalyzed PPMI, ($n = 4075$; $M_{age} = 64.9$, $M_{education} = 16.3$, 54.8% men, 95% White) selecting a mixed cohort sample at baseline. Confirmatory factor analysis tested the internal consistency and structural validity of their PDCC. Unfortunately, it had inadequate reliability ($\omega = 0.64$) and structural validity on all five fit indices (Table 1). We explored whether a differently weighted (vs. equally weighted) PDCC performed better and found adequate reliability ($\omega = 0.78$) with improved fit, although still inadequate structural validity. As the authors note, assigning weights can be vexing, but the reliability and validity of weighted PACCs is typically evaluated before deciding to unweight.[2,4,5] We demonstrate that Weintraub and colleagues'[1] PDCC does not meet select psychometric criteria and we caution its further use without additional information.

A sensitive and predictive PDCC is needed, however, we must incorporate lessons learned in AD research to strengthen methodology and appreciate the unique cognitive trajectories, motor influences, and clinical trial needs of PD.

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TABLE 1 *Confirmatory factor analysis fit indices for Weintraub and colleagues'[1] cognitive composite and modification*

	χ^2	χ^2 *P*	SRMR	RMSEA	CFI	TLI
Weintraub composite (equal weights)	4319.81	<0.001	0.226	0.282	0.203	0.146
Weintraub composite (different weights)	787.29	<0.001	0.088	0.146	0.863	0.772

Note: Weintraub composite is based on six tasks with equal weighting: Letter-Number Sequencing, Symbol Digit Modalities, Category Fluency, Benton Judgement of Line Orientation, Hopkins Verbal Learning Test Total Learning and Free Recall. Models were run in *lavaan* (0.6–19) with standardized scores (based on published reference groups) and robust Satorra-Bentler maximum likelihood estimation. All baseline PPMI cohorts included in analysis since an index should function well in both clinical and control participants. Inadequate model fit determined by χ^2 $P < 0.05$, SRMR >0.08, RMSEA >0.10, CFI < 0.90, TLI < 0.90. Analysis script is saved on the following repository: https://osf.io/h3a9g/?view_only=37c4191f262745ec9557d4e59d2dd1b8.

Abbreviations: SRMR, standardized root mean residual; RMSEA, root mean squared error of approximation; CFI, comparative fit index; TLI, Tucker-Lewis index; PPMI, Parkinson's Progression Markers Initiative.

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Data Availability Statement

The data that support the findings of this study are openly available in PPMI at https://www.ppmi-info.org/.

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